

Multimodal Fusion for Remote Sensing Land Cover Segmentation

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I. LITERATURE SURVEY

The literature survey focuses on multimodal embedding approaches, with an emphasis on their use in semantic segmentation. Recent studies have shown that incorporating textual information can improve segmentation performance by providing additional semantic context. TAMS adopts a UNetFormer-based architecture with a BERT text encoder and integrates visual and textual features using cross-attention and contrastive learning. This approach significantly enhances segmentation accuracy and remote sensing capabilities [1]. Similarly, FSVLM combines a segmentation model with a vision-language framework. It demonstrates that textual descriptions help capture spatial and contextual relationships that are difficult to model using visual features alone [2]. CLIP2RS builds on pretrained CLIP representations and introduces a dual-granularity alignment strategy that aligns both image-level and pixel-level features. This enables the effective use of vision-language representations for semantic segmentation and improves semantic consistency [3]. RemoteCLIP extends vision-language modeling to the remote sensing domain by constructing a large-scale image-text dataset. It learns domain-adapted multimodal representations that generalize well across different downstream tasks [4].

These findings highlight the importance of multimodal embeddings for improving representation quality in remote sensing applications. However, existing work primarily incorporates textual descriptions as fixed inputs, focusing on multimodal integration rather than the characteristics of the textual information itself. As a result, the influence of different forms of textual descriptions, such as variations in source, structure, or level of detail, on segmentation performance remains unclear. Furthermore, while textual guidance provides high-level semantic context, it does not explicitly represent the underlying class distribution within the image. These observations motivate the proposed approach.

II. PROJECT PROPOSAL

To address the identified limitations, this study proposes a multimodal semantic segmentation approach that examines the contribution of textual information to model performance. The research is guided by three questions: (i) whether incorporating textual descriptions improves segmentation performance compared to a vision-only baseline, (ii) how different types of textual descriptions (text-only, vision-based, and hybrid) affect segmentation performance, and (iii) whether incorporating ground truth class distribution as an auxiliary objective provides additional benefits beyond textual guidance. To answer these questions, a UNetFormer-based architecture is adopted as the visual backbone, and textual features are extracted using a pretrained vision-language model (CLIP). These textual embeddings are integrated into the segmentation model through text-visual cross-attention modules applied at multiple decoder stages. The proposed approach follows a controlled experimental design. It

compares an image-only baseline with multiple multimodal variants using different caption types, and further evaluating the effect of incorporating ground truth class distribution as an auxiliary objective implemented as a KL-divergence loss. Model performance will be assessed using mean Intersection over Union, pixel accuracy, and F1-score. This setup enables a systematic and controlled evaluation of multimodal integration and structured information.

III. PROOF OF CONCEPT

A proof of concept was conducted to evaluate the impact of incorporating textual descriptions on segmentation performance. The dataset was prepared by aligning images, segmentation masks, and captions, followed by exploratory analysis of class distribution and caption characteristics. An image-only UNetFormer baseline was trained and evaluated on a validation set. To assess the contribution of textual information, a multimodal variant of the model was implemented by integrating a pretrained CLIP text encoder with UNetFormer through text-visual cross-attention modules applied at multiple decoder stages. In this proof of concept, a single hybrid caption type was used as the textual input, and the text encoder was kept frozen while only the cross-attention layers were trained. The multimodal model demonstrated a consistent improvement in performance compared to the image-only baseline, resulting in a relative mIoU improvement of approximately 9%. These results indicate that textual descriptions provide useful semantic guidance that complements visual features in segmentation.

Future work will extend the current approach to address the remaining research questions. Planned improvements include the application of data augmentation techniques to increase data diversity and robustness, the use of domain-adapted vision-language models to improve cross-modal representation alignment, and extended training schedules to enable better convergence. In addition, more advanced text-vision interaction mechanisms will be explored to enhance the effectiveness of multimodal feature integration.

IV. REFERENCES

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